Lemma Chain: Collapse Stability for Recursive Scaffolded Navier–Stokes

Lemma 1: Boundedness of Scaffold Field $\psi(x,t)$

Statement: Let ψ evolve under:

$$\partial_t \psi + (u \cdot \nabla)\psi = -\alpha\psi + \nabla \cdot \mathcal{T}(u),$$

with $\alpha > 0$, and $\mathcal{T}(u) \in L^2$. Then for all $t > 0$, $\|\psi(t)\|_{L^2} \leq C$ for some finite constant C depending on initial data.

Proof Sketch: Multiply both sides by ψ , integrate:

$$\frac{1}{2}\frac{d}{dt}\|\psi\|^2 + \alpha\|\psi\|^2 = \int \nabla \cdot \mathcal{T}(u) \cdot \psi \, dx \leq \|\mathcal{T}(u)\| \cdot \|\nabla\psi\|$$

Use Young's inequality and Grönwall to show $\|\psi\|$ remains bounded.

Lemma 2: Energy Bound of Full System

Statement: For total energy

$$E(t) = \frac{1}{2}\int |u|^2 + \lambda|\psi|^2 \, dx$$

there exists $C > 0$ such that $E(t) \leq C$ for all $t \geq 0$.

Proof Sketch: Add Navier–Stokes and ψ equations, use inner products and divergence-free property to show:

$$\frac{d}{dt}E(t) \leq -\nu\|\nabla u\|^2 - \alpha\|\psi\|^2 + (\text{controlled transfer terms})$$

Collapse terms bounded by prior lemma → decay or steady-state convergence.

Lemma 3: Uniform Control of $\|\nabla u\|_\infty$

Statement: There exists $\delta > 0$ such that

$$\|\nabla u(t)\|_\infty \leq C + \delta\|\psi(t)\|_{H^1}$$

Proof Sketch: Use structure of $\mathcal{T}(u)$ to suppress nonlinearity. Bootstrap ∇u through vorticity identity:

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega + \nabla \times \psi$$

Bound each term; $\nabla \times \psi$ acts as strain-regulator.

Lemma 4: Classical Limit Preservation

Statement: As $\psi \to 0$ in L^2 , solution $u \to u_{NS}$, where u_{NS} satisfies the classical Navier–Stokes equations and remains smooth.

Proof Sketch: - Scaffold terms vanish uniformly in energy norms - Limit passes through weak formulation - Smoothness maintained via energy bounds and continuity in time

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These lemmas form the backbone of recursive collapse control over the NS system. Combined, they imply global regularity under scaffolded dynamics, with the classical system preserved in the limit.